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Stress-testing US colorectal cancer screening guidelines: Decennial colonoscopy from age 45 is robust to natural history uncertainty and colonoscopy sensitivity assumptions

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Abstract

Purpose. The 2023 American College of Physicians (ACP) guidelines for colorectal cancer (CRC) screening are at odds with the United States Preventive Task Force (USPSTF) guidelines, with the former recommending screening starting at age 50 y and the latter at age 45 y. This article "stress tests" CRC colonoscopy screening strategies to investigate their robustness to uncertainties stemming from the natural history of disease and sensitivity of colonoscopy.

Methods. This study uses the CRC-SPIN microsimulation model to project the life-years gained (LYG) under several colonoscopy CRC screening strategies. The model was extended to include birth cohort effects on adenoma risk. We estimated natural history parameters under 2 different assumptions about the youngest age of adenoma initiation. For each, we generated 500 parameter sets to reflect uncertainty in the natural history parameters. We simulated 26 colonoscopy screening strategies and examined 4 different colonoscopy sensitivity assumptions, encompassing the range of sensitivities consistent with prior tandem colonoscopy studies. Across this set of scenarios, we identify efficient screening strategies and report posterior credible

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intervals for benefits of screening (LYG), burden (number of colonoscopies), and incremental burden-effectiveness ratios.

Results. Projected absolute screening benefits varied widely based on assumptions, but strategies starting at age 45 y were consistently in the efficiency frontier. Strategies in which screening starts at age 50 y with 10-y intervals were never efficient, saving fewer life-years than starting screening at age 45 y and performing colonoscopies every 15 y while requiring more colonoscopies per person.

Conclusions. Decennial colonoscopy screening initiation at age 45 y remained a robust recommendation. Colonoscopy screening with a 10-y interval starting at age 50 y did not result in an efficient use of colonoscopies in any of the scenarios evaluated.

Keywords

cancer screening; colorectal cancer; robust decision making

Colorectal cancer (CRC) is the second-leading cause of cancer death in the United States. In 2023, 153,020 new CRC cases and 52,550 CRC deaths were projected in the United States.[1] From 1991 through 2018, the overall cancer death rates fell by 31%, a consistent decline attributed to smoking reduction, early detection, and improved treatment. For CRC specifically, the decrease in annual mortality from 1980 through 2018 was 53% among males (from 32.8 to 15.8 annual deaths per 100,000 people) and 55% among females (from 24.4 to 10.9 annual deaths per 100,000 people).[2]

Despite these significant overall decreases in CRC mortality, the age-specific incidence increased among adults younger than 50 y in the United States and other high-income countries.[3] The proportion of CRC cases among people younger than 55 y increased from 11% in 1995 to 20% in 2019.[1] Although early-onset CRC risk remains low in absolute terms, adults born around 1990 have double the risk of colon cancer and quadruple the risk of rectal cancer compared with those born in 1950.[4] Moreover, an increased incidence of CRC has been observed among people born since the 1950s worldwide.[5] The potential causes of this phenomenon are still unclear. Recent evidence suggests an association between early-onset CRC and alcohol consumption,[6] smoking,[7] and body composition and metabolism.[8]

Although it is clear that early-onset CRC is increasing,[3] there is less consensus about whether and how screening practices should be updated in response, which is evidenced by conflicting screening guidelines. The United States Preventive Task Force (USPSTF) reduced the recommended age to start screening from 50 to 45 y[9] and recommends multiple options for screening based on modeling evidence. In contrast, the American College of Physicians (ACP) discounted observational and modeling evidence and recommended CRC screening beginning between ages 50 to 75 y, with either decennial colonoscopy or biennial fecal immunochemical test.[10,11]

Cancer Incidence and Survival Modeling Network (CISNET) models used to inform policy simulate the CRC natural disease process, including initiation of adenomas, which are CRC precursor lesions; the transition of adenomas to asymptomatic, preclinical CRC; and the

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transition to clinically detected CRC and, ultimately, survival outcomes while accounting for death from other causes.[9] The models simulate the natural history of CRC in an average-risk, unscreened population. Then, the same population is simulated to undergo screening using a variety of screening modalities. Modeling the natural history of an unobserved process such as CRC progression requires assumptions, and up to this point, all CISNET colon models have assumed that adenomas can be initiated only at age 20 y or older[12] due to low prevalence and resulting scarcity of data in average-risk younger populations.[13] These assumptions cannot be verified with the available data; hence, quantifying their effect on screening benefit projections and recommendations is important, especially in the context of increasing early-onset CRC.

This article conducts a "stress test" of ACP's and USPSTF's colonoscopy screening guidelines by evaluating the robustness of colonoscopy screening strategies to natural history and test sensitivity uncertainties. First, we quantify natural history uncertainty by performing Bayesian calibration[14] of 2 specifications of the CRC-SPIN model. The first model specification allows adenomas to be initiated after age 20 y (baseline assumption consistent with recent analyses),[12] and the second specification reduces the minimum age of adenoma initiation to 10 y. After specifying both models, we calibrate them to represent the average-risk US population. Then, we simulate all colonoscopy screening strategies evaluated in the burden-effectiveness analysis that informed the most recent USPSTF screening recommendations[9] to assess how screening burden and effectiveness depend on uncertainty within and between model specifications. Finally, we investigate whether current colonoscopy screening recommendations are robust to uncertainties surrounding the natural history of CRC and the sensitivity of colonoscopy by examining the stability of the burden-effectiveness frontier.

Methods

Natural History Model

This article uses the Colorectal Cancer Simulated Population model for Incidence and Natural History (CRC-SPIN) to simulate the natural history and progression of adenomas to CRC. The model generates adenomas in a population of individuals using a nonhomogenous Poisson process. The model then simulates the growth of adenomas, progression to CRC, and downstream outcomes such as time at CRC stages and life-years, considering other-cause mortality. Colonoscopy yields benefits by removing adenomas before they progress to cancer to prevent CRC and detecting and removing CRC at earlier stages. CRC-SPIN has been described elsewhere,[14,15] extensively validated,[16–18] and used to inform cancer screening policies.[9,19]

CRC-SPIN simulates adenoma risk using a nonhomogenous Poisson process, detailed in Supplementary Appendix I. Individual-level risk of developing an adenoma depends on sex and age. This article introduces 2 modifications to the prior CRC-SPIN version,[15] which assumed no risk of developing an adenoma before age 20 y. First, we allowed adenomas to initiate at earlier ages, after age 10 y. Second, we allowed for increased adenoma initiation risk based on the birth year of each individual.

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The birth cohort effects model allows adenoma risk to vary for each birth cohort year, using a piecewise model with knots at years 1875, 1910, 1940, 1955, 1970, and 1975. The year 1940 is a reference point at which the effect is zero. The minimum and maximum years were chosen based on the birth cohorts of populations included in available calibration targets. The intermediate knots were chosen such that there is a finer resolution toward 1970, where the birth cohort effects will be more relevant and where risk changes were known to be more pronounced.[20] We produce a smooth, monotonic interpolation between these knots to define yearly birth cohort effects, using a method based on piecewise radial functions.[21,22] This effect captures secular changes that resulted in higher adenoma risk while making no additional assumptions about its underlying causes. This flexible functional form can accommodate the nonmonotonic relationship between CRC birth cohort effects and birth year observed in the United States.[5,20]

The birth cohort model we use is analogous to the approach used to adjust the adenoma risk to different populations seen in comparable modeling studies[9,18] and assumes that 1) increased CRC incidence risk is attributed to higher risk of adenoma initiation (as opposed to faster progression) and 2) the adenoma onset incidence risk ratio for a given birth cohort relative to the 1940 cohort is constant over their lifetime; that is, it does not vary with age. This structural assumption is consistent with the hypothesis that there is a cohort effect operating across ages that is not being reflected in older age groups due to the preventive effect of screening.[23] This assumption does not imply that the CRC incidence risk ratios of different cohorts relative to 1940 are constant conditional on age because CRC incidence also depends nonlinearly on age (Supplementary Appendix I). Supplementary Appendix I presents the natural history model in more detail, and Supplementary Table 1 presents prior distributions used in the analysis.

Bayesian Estimation of Natural History Parameters

Approximate Bayesian computation (ABC) is a framework that requires the specification of 1) calibration targets defined as summary statistics derived from data, 2) prior distributions for unknown (calibrated) parameters, and 3) a model that maps parameters to model-predicted targets. We use the incremental mixture approximate Bayesian computation algorithm[24] to estimate natural history model parameters and quantify uncertainty by estimating their joint posterior distribution. Supplementary Table 1 contains the prior distribution for each parameter used across both model specifications.

Calibration targets. This analysis uses a set of 43 calibration targets derived from 8 sources to calibrate CRC-SPIN to the average-risk US population. Following prior CRC-SPIN calibrations,[15,24] we use adenoma prevalence data[25] to inform adenoma risk by sex and age. We use targets derived from a computed tomographic colonoscopy study[26] to inform the distribution of adenoma size and 2 other observational studies[27,28] to inform the adenoma size at transition to CRC. The UK Flexible Sigmoidoscopy Screening Trial[29] informs the prevalence of preclinical CRC by sex. Consistent with prior modeling studies,[9] we use age-specific cancer incidence data by sex and location (rectal v. colon) from the Surveillance, Epidemiology, and End Results (SEER) 1975–1979 data set[30] to calibrate cancer incidence risk in the unscreened US population. This analysis included 2 new targets

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relative to prior CRC-SPIN calibrations.[15,24] Increased risk among new cohorts is informed by recent SEER data, using SEER 2009 and SEER 2014 age-specific incidence for young adults (aged 40–44 y).[30] Supplementary Table 2 lists each calibration target and their tolerance intervals.

Model specifications. So far, CISNET-CRC models allow adenoma initiation only after age 20 y,[12] partly due to low prevalence and scarcity of data in average-risk younger populations.[13] This assumption may affect the estimation of disease progression parameters. Assuming later adenoma onset might require faster disease progression for adenoma prevalence targets to be matched at younger ages and might also imply stronger adenoma risk birth cohort effects to match CRC risk observed in the US population.[4] Disease progression parameters are a crucial feature of CRC models[31]; hence, ensuring that policy recommendations are robust to this assumption is important.

This article uses 2 specifications of the CRC-SPIN model to investigate how the minimum age of adenoma initiation assumption affects model estimates and downstream screening effectiveness projections. In the baseline model specification (BC-20), we set the minimum age at adenoma initiation to 20 y, consistent with prior CRC-SPIN analyses. In a second model specification (BC-10), we set the minimum age to 10 y. Increasing cancer incidence at younger ages[4] suggests that the BC-20 specification is less plausible than BC-10. This article investigates the importance of this model assumption by calibrating and evaluating screening policies for both model specifications.

CRC Screening Experiments

Colonoscopy screening strategies. This article focuses on evaluating screening colonoscopy, considering that it is the most-used screening test in the United States. We evaluated the 26 colonoscopy screening strategies previously considered in a decision analysis supporting the USPSTF CRC screening recommendations.[9] These strategies vary in 3 policy levers: age to start screening (45, 50, 55 y), periodicity of screening colonoscopies (5, 10, 15 y), and age to end screening (70, 75, 80, 85 y). A grid experimental design over the 3 policy levers results in 36 strategies, but only 26 are simulated after removing redundant combinations. Screening strategies are coded as [age to start]2[age to end], [interval]. For instance, strategy 45–75,10 refers to performing colonoscopy screening at ages 45, 55, 65, and 75 y. Individuals with an adenoma detected during colonoscopy screening undergo colonoscopy surveillance following the Multi-Society Task Force guidelines.[32]

These strategies encompass all current US CRC colonoscopy screening recommendations. Strategy 45–75,10 represents the current USPSTF guideline[33] and the ACG colonoscopy guideline,[34] whereas strategy 50–70,10 represents the current ACP guideline[11] for colonoscopy (ACP's guideline recommends colonoscopy screening from age 50–75 every 10 y, but it would be done only at ages 50, 60, and 70 y within our model representing full adherence to guidelines). Consistent with comparable studies that inform guidelines,[9] we simulate full adherence to screening. Thus, our estimates focus on the effect of screening

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on those who follow screening guidelines as opposed to the population-average effect of screening invitations.

Colonoscopy sensitivity. The sensitivity of colonoscopy examinations is uncertain and heterogeneous across gastroenterologists and clinics.[35] Colonoscopy is fundamentally an operator-dependent procedure, and higher adenoma detection rates are associated with lower rates of postcolonoscopy CRC.[36] Hence, assessing the extent to which colonoscopy sensitivity affects screening effectiveness and the efficiency of screening strategies is crucial.

We consider 4 colonoscopy sensitivity scenarios varying sensitivity to the detection of adenomas: high, baseline, low, and very low (Supplementary Appendix II, Supplementary Table 3). The baseline scenario matches the sensitivity assumption used in the analyses that informed the current USPSTF CRC screening guidelines.[9] The high- and low-colonoscopy-sensitivity scenarios match assumptions used in prior sensitivity analyses that informed guidelines.[19,37] Further, the very-low-sensitivity scenario reflects evidence that baseline colonoscopy sensitivity assumptions may be too optimistic.[16] Supplementary Appendix II demonstrates that tandem colonoscopy studies do not fully identify colonoscopy sensitivity (especially for small adenomas that are more likely to be missed) and demonstrates that all scenarios we consider in this analysis are plausible by reanalyzing data from a meta-analysis of tandem colonoscopy studies (Supplementary Figure 1).[38]

Sensitivity to CRC and lesion size is handled as follows. Before the adenoma progresses to CRC, lesion size determines the sensitivity of colonoscopy to detect and remove the lesion, according to the scenarios discussed in Supplementary Appendix II. After progression to CRC, we use the maximum of 2 values: the lesion size–based sensitivity (also used only for adenomas) and CRC sensitivity. Following the approach used in comparable analyses,[9] the (minimum) CRC sensitivity is set at 0.91, which is 1 – the adenoma detection rate estimated for large (≥10 mm) adenomas in a tandem colonoscopy study meta-analysis (0.91).[39] This analysis assumes colonoscopy sensitivity is constant over time conditional on lesion size and status and does not vary across individuals.

Outcomes. This study evaluates health outcomes in a simulated cohort of 5 million average-risk US adults, starting at age 40 y and followed until death. Consistent with recent analyses used to guide policy,[9] we use life-years gained (LYG) from screening as the effectiveness measure by comparing a no-screening scenario to each scenario in which individuals are screened. The number of colonoscopies (n_{col}) performed is used as the burden measure. The incremental burden-effectiveness ratios (IBERs) are computed by comparing 2 nondominated screening strategies as $IBER = \Delta n_{col}/\Delta LYG$, where Δn_{col} is the additional number of colonoscopies required by a more intensive screening regimen and ΔLYG is the incremental number of LYG by that regimen. Also consistent with prior analyses,[9] we do not use incremental cost-effectiveness ratios because this analysis does not consider costs.

Experimental design. Following the Robust Decision-Making decision-analytic approach,[40] we created a large-scale experimental design to stress test the efficiency of alternative screening strategies. The experimental design of this study considered natural

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history uncertainty (represented by the posterior distribution of model parameters), structural assumptions (represented by 2 different model specifications), and uncertainty related to the efficacy of the interventions (represented by the 4 sensitivity scenarios). Hence, the experimental design of this study consisted of the combination of 2 model specifications, 500 natural history parameter sets for each model specification, 26 screening strategies, 1 "no-screening" scenario used as the comparator, and 4 colonoscopy sensitivity scenarios, resulting in 105,000 unique model runs. The dimensions of the experimental design were defined such that the screening experiments would be performed using a computing budget of approximately 250,000 core hours. Each unique model run simulates 5 million individuals. This experimental design simulates 525 billion life histories and more than 1.5 trillion adenomas.

Burden-Effectiveness Frontier

We determine the efficiency status of each strategy within each scenario, formed by combining 4 colonoscopy sensitivity scenarios and 2 model specifications. Within each of the 8 scenarios, each of the 26 strategies can either be 1) efficient (i.e., nondominated), 2) extended dominated, or 3) dominated. Efficient strategies are those for which there is no alternative strategy (or combination of strategies) that results in higher LYG requiring a lower number of colonoscopies, and the burden-effectiveness frontier is formed by only nondominated strategies. Extended-dominated strategies are not dominated by any single strategy but by a combination of 2 efficient strategies. All other strategies are dominated.[41]

Robustness of efficiency status. First, we compute the efficiency status of all strategies by using the expected effectiveness and burden of the strategy (i.e., $E_{\theta}[Y]$, where θ_i is 1 of the natural history parameter sets obtained via Bayesian calibration). Once the efficiency status is determined, we compute the *IBER* for each θ_i, while holding the efficiency status constant, which ensures that the comparators are constant across the θ_i parameters and the *IBER* metric is meaningful. We then report 95% prediction intervals for IBER for each scenario.

However, the efficiency frontier is not the same across all θ_i parameters. A strategy that is efficient in expectation might be inefficient within many natural history parameters θ_i, which are consistent with the empirical evidence used to calibrate the model. Hence, we evaluate the robustness of the efficiency status of each strategy by computing the burden-effectiveness frontier for each θ_i (i.e., $E[Y|\theta_i]$, where the expectation is taken only across the population). Then, we evaluate the probability that each strategy is at the burden-effectiveness frontier for each parameter set. A strategy in the frontier across all parameter sets and all scenarios can be recommended with more confidence because it demonstrates that its efficiency is insensitive to any of the uncertainties considered in the study. In contrast, a strategy that is never on the efficiency frontier cannot be recommended on the grounds of the model.

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Results

Natural History Parameter Estimates

Birth cohort effects were nonmonotonic, with a sharp increase (and higher uncertainty) in the last birth year calibrated (Figure 1). Birth cohort effects were distant from 1 for most of the period analyzed, suggesting that the assumption of no birth cohort effects is implausible. Natural history parameters, their prior distribution used for calibration, and their respective 95% posterior credible intervals are reported separately for each model in Supplementary Table 1. While some parameter estimates are similar across models, model estimates are contingent on the assumed minimum age at adenoma initiation (Supplementary Figure 2). For instance, model BC-20 predicts a lower adenoma prevalence at age 25 y than model BC-10 does, and the posterior distributions of the models do not overlap. Unsurprisingly, the model that allows earlier adenoma initiation (BC-10) implies a higher cumulative risk of adenoma initiation by age 25 y. Despite those differences, both models have approximately the same implied cumulative adenoma initiation risk at age 80 y. While clinical expertise or future evidence might favor one model specification over the other, this study uses both models' posterior distributions without making further assumptions about how likely each model is to be closest to the real world.

Base-Case Burden-Effectiveness Estimates

In our base-case scenario (baseline colonoscopy sensitivity, model BC-20), the ACP strategy (50–70,10) provided 317 (254, 406) LYG/1,000 people and required 3,410 (3,300, 3,550) colonoscopies/1,000 people but was dominated by strategy 45–70,10, which provided 343 (275, 440) requiring 3,640 (3,510, 3,800) colonoscopies/1,000 people (Table 1). The USPSTF strategy (45–75,10) provided 347 (279, 444) LYG/1,000 people requiring 4,200 (4,110, 4,320) colonoscopies/1,000 people and was not dominated by any single strategy but was extended dominated by a combination of strategies 45–70,10 and 45–70,5. That said, the USPSTF strategy was within 3 days of the frontier, which would make it "nearly efficient" in the analysis that informed USPSTF guidelines and was in the frontier for model specification BC-10 (Figure 2). All strategies on the frontier that performed screening every 10 y or more frequently started at age 45 y. The only efficient strategy that began screening after age 45 y was strategy 55–70,15, which performed 2 screening tests 15 y apart.

Robustness Analyses

LYG estimates are highly contingent on sensitivity assumptions for all strategies (Supplementary Table 4, Supplementary Figure 3). Under baseline sensitivity and model assumptions (sensitivity = baseline and model = BC-20), strategy 45–75,10 results in 347 (279, 444) LYG/1,000 people (Supplementary Table 4). Under a high-sensitivity scenario, the same strategy results in 356 (288, 455) LYG/1,000 people. The benefit from the same policy decreases to 311 (248, 398) if sensitivity is very low. This effectiveness gap of 10.3 percentage points between the base-case scenario and the very-low-sensitivity scenario is similar to other strategies. For example, strategy 50–70,10 yields 11% fewer LYG under the very-low-sensitivity scenario relative to our baseline scenario.

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The number of colonoscopies performed is similar across sensitivity scenarios and natural history assumptions but is contingent on the intensity of the screening strategy (Supplementary Table 5). Under baseline assumptions, strategy 50–70,10 is expected to result in 3,410 (3,300, 3,550) colonoscopies per 1,000 individuals. This estimate increases to 3,640 (3,510, 3,800) for the 45–70,10 strategy and to 4,200 (4,110, 4,320) for the 45–75,10 strategy. For any particular strategy, there are slight differences in the number of colonoscopies by sensitivity scenario because higher sensitivity to adenoma detection can result in more people being referred to colonoscopy surveillance.

Burden-effectiveness ratios vary only slightly across colonoscopy sensitivity scenarios and more so across model specifications (Supplementary Table 6). Because burden-effectiveness ratios are calculated only for efficient strategies, it is useful to examine when they are not calculated. Notably, strategy 50–70,10 was never on the efficiency frontier in any scenario, and strategy 45–75,10 was in the frontier only under the high-sensitivity scenario or in the baseline sensitivity scenario for model BC-10 (Supplementary Figure 3). Strategy 45–70,10 was in the frontier in all scenarios, demonstrating that starting screening at age 45 y and ending at age 70 y (with the last screening colonoscopy at age 65 y) was a robust recommendation. However, extending screening to age 75 y (and thus, the fourth screening test) was not efficient across all scenarios. Strategy 45–75,10 had an IBER of 126 (96.4, 163) colonoscopies per LYG for baseline sensitivity assumptions (model BC-10) and 125 (95.3, 163) colonoscopies per LYG for high sensitivity (Supplementary Table 6). Starting decennial screening at age 45 y but ending at age 70 y required fewer colonoscopies per LYG: 40.2 (32.1, 49.4) for baseline sensitivity. Efficiency ratios were not computed for the ACP-recommended strategy because it was never efficient.

Robustness of efficiency status. Our previous results suggest that the USPSTF-recommended strategy (45–75,10) is not always on the efficiency frontier and that the ACP-recommended strategy (50–70,10) is never on the efficiency frontier (Figure 2, Supplementary Figure 3). However, those results are based on the strategies' expected effectiveness and burden, and those vary widely across parameter sets, as indicated by the wide uncertainty bounds in Figure 2. To measure the robustness of the efficiency status of each strategy, we analyzed the efficiency status for each strategy in each of the 500 parameter sets (within each model specification and colonoscopy sensitivity scenario). We computed the efficiency status for all strategies across each of those 4,000 scenarios (500 parameter sets × 2 model specifications × 4 sensitivity scenarios) to compute the probability that each strategy is in the burden-effectiveness frontier (Figure 3). A strategy with a high probability of being efficient can be recommended more confidently than one with a low probability of being efficient. The USPSTF-recommended strategy (45–75,10) is not always on the efficiency frontier, but when it is not, it is only extended dominated (i.e., dominated by a combination of strategies) and was never dominated by a single strategy. The USPSTF strategy is more likely to be efficient when high colonoscopy sensitivity is assumed. In contrast, the ACP-recommended strategies had zero probability of being efficient across all scenarios we analyzed. We also confirm that strategy (45–70,10) was always efficient across all scenarios (Figure 3).

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Discussion

This analysis does not support ACP's guidance that decennial colonoscopy screening should start at age 50 y (guidance statements 1 and 4b).[11] Across all scenarios simulated (4,000 scenarios combining 4 colonoscopy sensitivity scenarios, 500 parameter sets representing natural history uncertainty, and 2 model specifications), a decennial colonoscopy strategy that starts at age 50 y did not use colonoscopies efficiently. ACP CRC screening guidelines are less intensive than USPSTF guidelines are. Based on our results, efficient and less-intensive screening regimens include stopping screening at age 70 y (i.e., avoiding a fourth screening colonoscopy) or screening at ages 55 and 70 y.

LYG by colonoscopy varied substantially across and within scenarios, but their order of magnitude was consistent with the existing literature. For instance, estimated LYG for strategy 45–75,10 was 347 (279, 444) LYG/1,000 people for the specification BC-20 and 329 (275, 403) for specification BC-10. The range of LYG for the same strategies produced by the 3 CISNET models was 291 to 361.[9] The point estimate from our study falls within this range, and the range of outcomes in our study has a wide overlap with ranges previously estimated. Our uncertainty intervals reflect uncertainty in the natural history of parameters and birth cohort effects that are not incorporated in prior analyses.

The birth cohort effects profile we estimated for the United States were nonmonotonic with birth year and were higher than the one before and after 1940. This U-shaped pattern around 1940 might seem puzzling, yet this pattern is consistent with CRC incidence patterns observed in the United States.[5] The evidence suggests that CRC risk changes over time, which presents challenges for modelers. Although we abstain from attempting to model the mechanisms through which those patterns emerge, this article demonstrates the feasibility of characterizing uncertainty in birth cohort effects jointly with other natural history parameters in CRC models.

Prior modeling studies evaluated the importance of sensitivity of CRC screening tests to detect precursor CRC lesions,[42–45] but few assessed the effect of colonoscopy sensitivity assumptions on screening effectiveness.[19,42] The analysis that informed the 2016 USPSTF guidelines[19] evaluated the impact of colonoscopy sensitivity on estimates and included a best-case and worst-case scenario, finding a 1% to 3% LYG difference between the base-case colonoscopy assumptions and the best- and worst-case scenarios for a decennial colonoscopy strategy starting at age 50 y and ending at age 75 y.[19] Our article further probes the importance of this assumption by including a very-low-sensitivity scenario with worse sensitivity than Knudsen et al.[19] In our worst-case colonoscopy sensitivity scenario, we project LYG to be 11 percentage points lower than assumed in our base-case analysis. Yet, the main recommendation that decennial screening should start at age 45 y remains valid. These findings also highlight the importance of access to high-sensitivity screening to ensure that the benefits of screening are equitable. If access to high-sensitivity screening is unequal, then the benefits of screening will not be equally distributed in the population.

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Strengths and Limitations

This article is the first CRC microsimulation study to produce a joint posterior distribution of adenoma initiation birth cohort effects and natural history parameters and use it to assess the robustness of CRC screening recommendations. The birth cohort model used in this study offers one alternative to explain the increase in CRC incidence. This approach mirrors the prior approach used by CISNET models[9] but produces a posterior distribution reflecting uncertainty in birth cohort effect parameters.

This study also presents limitations and opportunities for further extensions of this work. First, we only present 1 explanation for increased CRC risk through an overall increase in the risk of adenoma initiation based on birth cohorts. We did not evaluate other plausible mechanisms for increased early onset, such as different disease progression or a potential interaction between age and birth cohort effects. Second, we did not consider costs, following the decision analysis that informed the most recent recommendations of the USPSTF.[9] Third, we do not include the sessile serrated pathway to CRC. We also evaluated only the colonoscopy screening strategies considered in the analysis that informed the USPSTF 2021 decision[9]; a broader set of strategies (tests, intervals, age to start and end screening) could be evaluated in future work. Future work could further investigate the significance of uncertainty by computing the opportunity cost associated with suboptimal screening strategies and performing value-of-information analyses.[46]

Conclusion

Our results provide further evidence on the robustness of recommending colonoscopy screening initiation at age 45 y. Colonoscopy screening with a 10-y interval starting at age 50 y did not result in an efficient use of colonoscopies in any of the scenarios we evaluated. Less-intensive strategies that were also efficient included stopping screening earlier (at age 70 y) or performing screening every 15 y, but performing decennial screening starting at age 50 y was projected to underperform. This result was consistent across 4,000 scenarios, combining 2 model specifications, 4 levels of colonoscopy sensitivity, and 500 parameter sets obtained via Bayesian model calibration.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Data Availability

Detailed documentation of the CISNET colorectal cancer models used in this study is publicly available at https://cisnet.cancer.gov/colorectal/. The source code of the CISNET models is not open source.

References

1. Siegel RL, Wagle NS, Cercek A, Smith RA, Jemal A. Colorectal cancer statistics, 2023. CA Cancer J Clin. 2023;73(3):233–54. Available from: DOI: 10.3322/caac.21772; https://acsjournals.onlinelibrary.wiley.com/doi/10.3322/caac.21772 [PubMed: 36856579]
2. National Cancer Institute. SEER Explorer—long term trends, 2021. Available from: https://seer.cancer.gov/explorer/application.html?site=20&data_type=2&graph_type=1&compareBy=sex&chk_sex_3=3&chk_sex_2=2&race=1&age_range=1&advopt_precision=1&advopt_show_ci=on&advopt_display=1
3. Sinicrope FA. Increasing incidence of early-onset colorectal cancer. N Engl J Med. 2022;386(16):1547–58. DOI: 10.1056/NEJMra2200869 [PubMed: 35443109]
4. Siegel RL, Fedewa SA, Anderson WF, et al. Colorectal cancer incidence patterns in the United States, 1974–2013. J Natl Cancer Inst. 2017;109(8):27–32. DOI: 10.1093/jnci/djw322
5. Murphy CC, Zaki TA. Changing epidemiology of colorectal cancer—birth cohort effects and emerging risk factors. Nat Rev Gastroenterol Hepatol. 2024;21(1):25–34. DOI: 10.1038/s41575-023-00841-9 [PubMed: 37723270]
6. Chen J, Zhang IL, Terry MB, Yang W. Dietary factors and early-onset colorectal cancer in the United States—an ecologic analysis. Cancer Epidemiol Biomarkers Prev. 2023;32(2):217–25. DOI: 10.1158/1055-9965.EPI-22-0442 [PubMed: 36129804]
7. Li H, Chen X, Hoffmeister M, Brenner H. Associations of smoking with early- and late-onset colorectal cancer. JNCI Cancer Spectr. 2023;7(1):pkad004. DOI: 10.1093/jncics/pkad004 [PubMed: 36759940]
8. Laskar R, Qu C, Huyghe J, et al. Genome-wide association study and Mendelian randomization analyses provide insights into the causes of early-onset colorectal cancer. Ann Oncol. 2024;35(6):523–36. DOI: 10.1016/j.annonc.2024.02.008 [PubMed: 38408508]
9. Knudsen AB, Rutter CM, Peterse EF, et al. Colorectal cancer screening: an updated modeling study for the US preventive services task force. JAMA. 2021;325(19): 1998–2011. DOI: 10.1001/jama.2021.5746 [PubMed: 34003219]
10. Bretthauer M, Yang YX. New American college of physicians guidance on colorectal cancer screening: less is more. Ann Intern Med. 2023;176(8):1127–8. DOI: 10.7326/M23-1695 [PubMed: 37523706]
11. Qaseem A, Harrod CS, Crandall CJ, et al. Screening for colorectal cancer in asymptomatic average-risk adults: a guidance statement from the American College of Physicians (version 2). Ann Intern Med. 2023;176(8):1092–100. DOI: 10.7326/M23-0779 [PubMed: 37523709]
12. Knudsen AB, Rutter CM, Peterse EFP, et al. Colorectal Cancer Screening: An Updated Decision Analysis for the U.S. Preventive Services Task Force. U.S. Preventive Services Task Force Evidence Syntheses, formerly Systematic Evidence Reviews. Rockville (MD): Agency for Healthcare Research and Quality; 2021. Available from: http://www.ncbi.nlm.nih.gov/books/NBK570833/
13. Wong MCS, Huang J, Huang JLW, et al. Global prevalence of colorectal neoplasia: a systematic review and meta-analysis. Clin Gastroenterol Hepatol. 2020;18(3): 553–561.e10. DOI: 10.1016/j.cgh.2019.07.016 [PubMed: 31323383]
14. Rutter CM, Savarino JE. An evidence-based microsimulation model for colorectal cancer: validation and application. Cancer Epidemiol Biomarkers Prev. 2010;19(8): 1992–2002. DOI: 10.1158/1055-9965.EPI-09-0954 [PubMed: 20647403]

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15. DeYoreo M, Rutter CM, Ozik J, Collier N. Sequentially calibrating a Bayesian microsimulation model to incorporate new information and assumptions. BMC Med Inform Decis Mak. 2022;22(1):12. DOI: 10.1186/s12911-021-01726-0 [PubMed: 35022005]
16. Rutter CM, Nascimento de Lima P, Lee JK, Ozik J. Too good to be true? Evaluation of colonoscopy sensitivity assumptions used in policy models. Cancer Epidemiol Biomarkers Prev. 2022;31(4):775–82. DOI: 10.1158/1055-9965.EPI-21-1001 [PubMed: 34906968]
17. Rutter CM, Knudsen AB, Marsh TL, et al. Validation of models used to inform colorectal cancer screening guidelines. Med Decis Making. 2016;36(5):604–14. DOI: 10.1177/0272989X15622642 [PubMed: 26746432]
18. Van Den Berg DM, Nascimento de Lima P, Knudsen AB, et al. NordICC trial results in line with expected colorectal cancer mortality reduction after colonoscopy: a modeling study. Gastroenterology. 2023;165(4):1077–9.e2. DOI: 10.1053/j.gastro.2023.06.035 [PubMed: 37454978]
19. Knudsen AB, Zauber AG, Rutter CM, et al. Estimation of benefits, burden, and harms of colorectal cancer screening strategies: modeling study for the US preventive services Task Force. JAMA. 2016;315(23):2595–609. DOI: 10.1001/jama.2016.6828 [PubMed: 27305518]
20. DeYoreo M, Rutter CM, Lee SD. Use of 2-stage modeling to identify how colorectal cancer risk changes with period and cohort. Am J Epidemiol. 2023;192(2):230–36. DOI: 10.1093/aje/kwac177 [PubMed: 36222654]
21. Stineman RW. A consistently well behaved method of interpolation. Creat Comput. 1980;6(7):54–7.
22. Jhannesson T, Bjornsson H, Grothendieck G. stinepack: stineman, a Consistently Well Behaved Method of Interpolation. 2018. Available from: https://cran.r-project.org/web/packages/stinepack/stinepack.pdf
23. Neugut AI, Lebwohl B. Early onset colorectal cancer: a hypothesis. Oncologist. 2023;28(12):1015–6. DOI: 10.1093/oncolo/oyad274 [PubMed: 37758481]
24. Rutter CM, Ozik J, DeYoreo M, Collier N. Microsimulation model calibration using incremental mixture approximate Bayesian computation. Ann Appl Stat. 2019;13(4): 2189–212. DOI: 10.1214/19-AOAS1279 [PubMed: 34691351]
25. Corley DA, Jensen CD, Marks AR, et al. Variation of adenoma prevalence by age, sex, race, and colon location in a large population: implications for screening and quality programs. Clin Gastroenterol Hepatol. 2013;11(2):172–80. DOI: 10.1016/j.cgh.2012.09.010 [PubMed: 22985608]
26. Pickhardt PJ, Choi JR, Hwang I, et al. Computed tomographic virtual colonoscopy to screen for colorectal neoplasia in asymptomatic adults. N Engl J Med. 2003;349(23): 2191–200. DOI: 10.1056/NEJMoa031618 [PubMed: 14657426]
27. Church JM. Clinical significance of small colorectal polyps. Dis Colon Rectum. 2004;47(4):481–85. DOI: 10.1007/s10350-003-0078-6 [PubMed: 14994108]
28. Lieberman D, Moravec M, Holub J, Michaels L, Eisen G. Polyp size and advanced histology in patients undergoing colonoscopy screening: implications for CT colonography. Gastroenterology. 2008;135(4):1100–5. DOI: 10.1053/j.gastro.2008.06.083 [PubMed: 18691580]
29. Atkin WS, Edwards R, Kralj-Hans I, et al. Once-only flexible sigmoidoscopy screening in prevention of colorectal cancer: a multicentre randomised controlled trial. Lancet. 2010;375(9726):1624–33. DOI: 10.1016/S0140-6736(10)60551-X [PubMed: 20430429]
30. National Cancer Institute. Surveillance, Epidemiology, and End Results (SEER) Program. Incidence - SEER research data with delay-adjustment, 9 registries, Nov 2020 sub (1975–2018). 2022. Available from: www.seer.cancer.gov
31. Kuntz KM, Lansdorp-Vogelaar I, Rutter CM, et al. A systematic comparison of microsimulation models of colorectal cancer: the role of assumptions about adenoma progression. Med Decis Making. 2011;31(4):530–39. DOI: 10.1177/0272989X11408730 [PubMed: 21673186]
32. Gupta S, Lieberman D, Anderson JC, et al. Recommendations for follow-up after colonoscopy and polypectomy: a consensus update by the US multi-society task force on colorectal cancer. Gastroenterology. 2020;158(4):1131–53.e5. Available from: DOI: 10.1053/j.gastro.2019.10.026 [PubMed: 32044092]

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33. US Preventive Services Task Force. Screening for colorectal cancer: US preventive services task force recommendation statement. JAMA. 2021;325(19):1965–77. DOI: 10.1001/jama.2021.6238 [PubMed: 34003218]
34. Shaukat A, Kahi CJ, Burke CA, Rabeneck L, Sauer BG, Rex DK. ACG clinical guidelines: colorectal cancer screening 2021. Am J Gastroenterol. 2021;116(3):458. DOI: 10.14309/ajg.0000000000001122 [PubMed: 33657038]
35. Corley DA, Jensen CD, Marks AR, et al. Adenoma detection rate and risk of colorectal cancer and death. N Engl J Med. 2014;370(14):1298–306. DOI: 10.1056/nejmoa1309086 [PubMed: 24693890]
36. Schottinger JE, Jensen CD, Ghai NR, et al. Association of physician adenoma detection rates with postcolonoscopy colorectal cancer. JAMA. 2022;327(21):2114–22. DOI: 10.1001/jama.2022.6644 [PubMed: 35670788]
37. Zauber A, Lansdorp-Vogelaar I, Knudsen AB, Wilschut J, van Ballegooijen M, Kuntz KM. *Annals of Internal Medicine* clinical guidelines evaluating test strategies for colorectal cancer screening. Ann Intern Med. 2008;149(9):659–69. [PubMed: 18838717]
38. Van Rijn JC, Reitsma JB, Stoker J, Bossuyt PM, van Deventer SJ, Dekker E. Polyp miss rate determined by tandem colonoscopy: a systematic review. Am J Gastroenterol. 2006;101(2):343–50. DOI: 10.1111/j.1572-0241.2006.00390.x [PubMed: 16454841]
39. Zhao S, Wang S, Pan P, et al. Magnitude, risk factors, and factors associated with adenoma miss rate of tandem colonoscopy: a systematic review and meta-analysis. Gastroenterology. 2019;156(6):1661–74.e11. DOI: 10.1053/j.gastro.2019.01.260 [PubMed: 30738046]
40. Lempert RJ, Groves DG, Popper SW, et al. A general, analytic method for generating robust strategies and narrative scenarios. Manage Sci. 2006;52(4):514–28. DOI: 10.1287/mnsc.1050.0472
41. Neumann EbPJ, Sanders GD, Russell LB, et al., eds. Cost-Effectiveness in Health and Medicine. 2nd ed. Oxford (UK): Oxford University Press; 2016.
42. Thiruvengadam NR, Coté GA, Gupta S, et al. An evaluation of critical factors for the cost-effectiveness of real-time computer-aided detection: sensitivity and threshold analyses using a microsimulation model. Gastroenterology. 2023;164(6):906–20. DOI: 10.1053/j.gastro.2023.01.027 [PubMed: 36736437]
43. Deibel A, Deng L, Cheng CY, et al. Evaluating key characteristics of ideal colorectal cancer screening modalities: the microsimulation approach. Gastrointest Endosc. 2021;94(2):379–90.e7. DOI: 10.1016/j.gie.2021.02.013 [PubMed: 33600806]
44. Ladabaum U, Mannalithara A, Weng Y, et al. Comparative effectiveness and cost-effectiveness of colorectal cancer screening with blood-based biomarkers (liquid biopsy) vs fecal tests or colonoscopy. Gastroenterology. 2024;167(2): 378–91. DOI: 10.1053/j.gastro.2024.03.011 [PubMed: 38552670]
45. Nascimento de Lima P, Van Den Puttelaar R, Knudsen AB, et al. Characteristics of a cost-effective blood test for colorectal cancer screening. J Natl Cancer Inst. 2024; 116(10):1612–20. DOI: 10.1093/jnci/djae124 [PubMed: 38845072]
46. Claxton KP, Sculpher MJ. Using value of information analysis to prioritise health research. PharmacoEconomics. 2006;24(11):1055–68. DOI: 10.2165/00019053-200624110-00003 [PubMed: 17067191]

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Highlights

- Colorectal cancer colonoscopy screening strategies initiated at age 45 y were projected to yield more life-years gained while requiring the least number of colonoscopies across different model assumptions about disease natural history and colonoscopy sensitivity.
- Colonoscopy screening starting at age 50 y with a 10-y interval consistently underperformed strategies that started at age 45 y.

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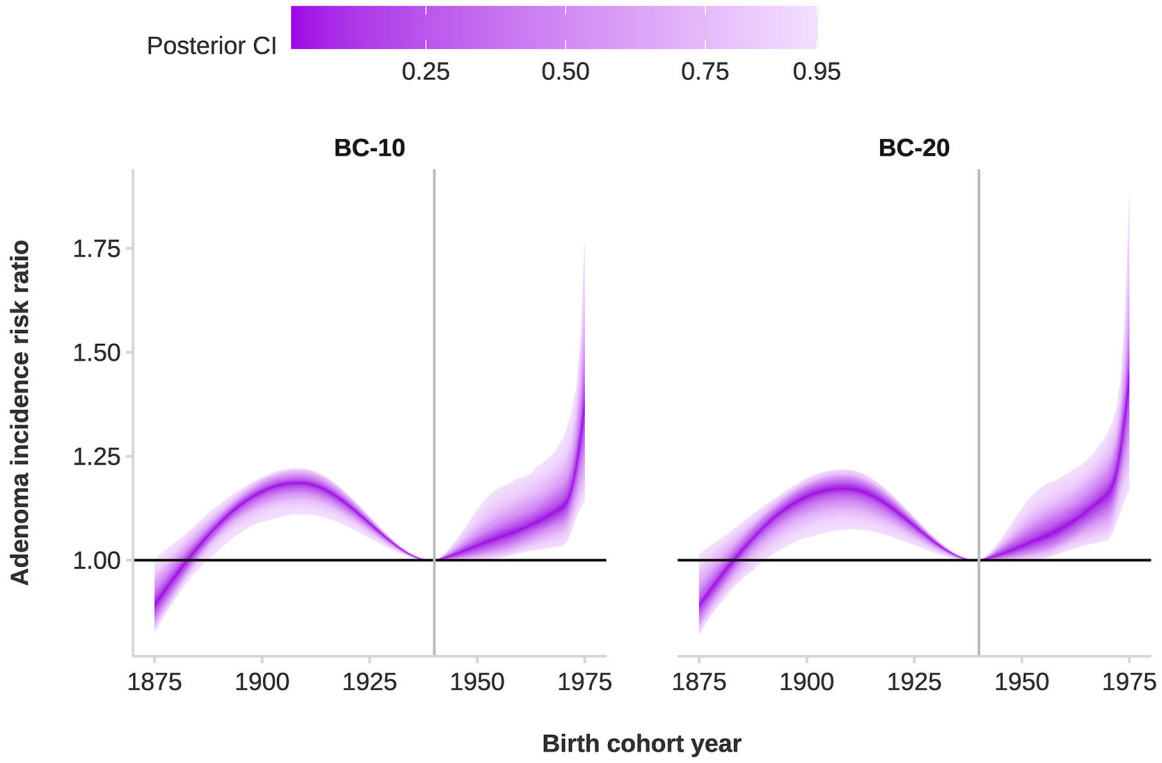


Figure 1.

Birth cohort incidence risk ratio estimates by model.

Shaded areas represent the posterior distribution of adenoma incidence risk ratios by cohort year and model. A gray vertical line crosses the 1940 birth cohort, used as a reference cohort. The horizontal black line at 1 represents the hypothesis of no change in adenoma initiation risk by birth cohort. The vertical gray line, set at 1940, represents the reference cohort against which all other cohorts are compared. The 2 panels present results for the 2 models calibrated independently to the same data. BC-10 and BC-20 refer to the models with a minimum age at adenoma initiation of 10 and 20 y, respectively.

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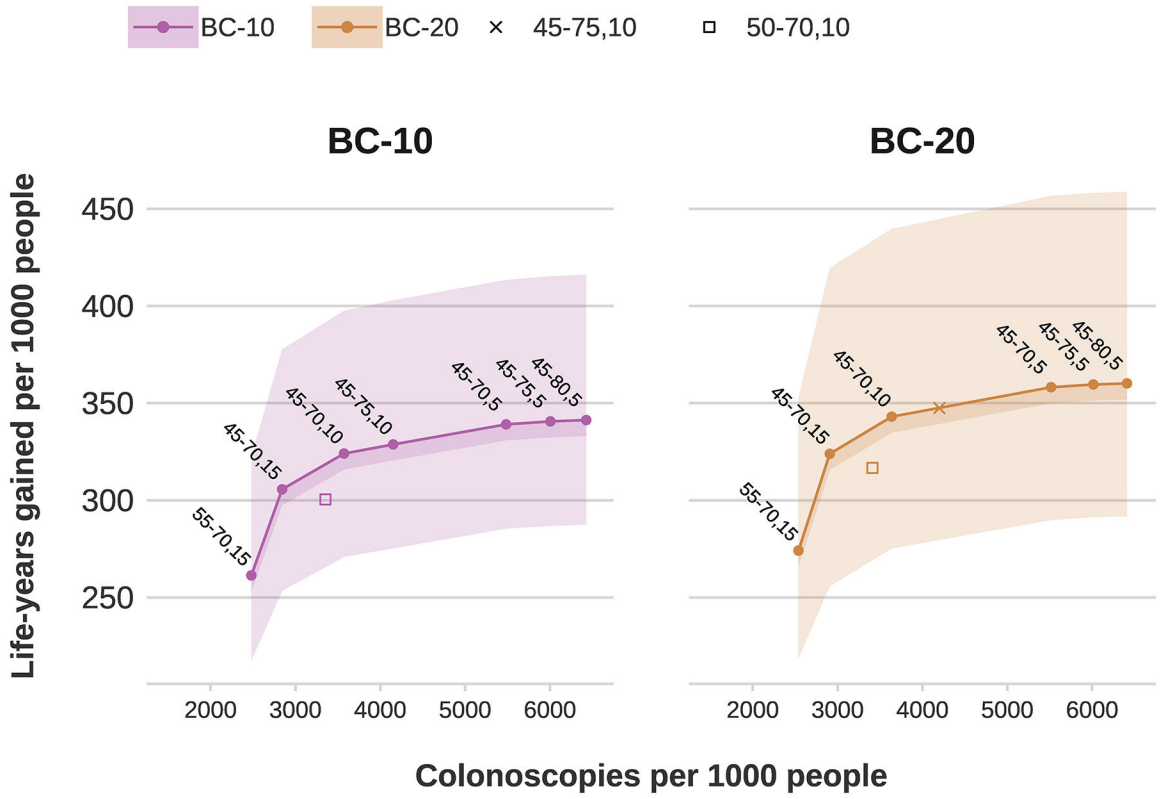


Figure 2.

Efficient screening strategies across model specifications.

Each symbol represents 1 screening strategy, and the line connecting them approximates the efficient frontier. Shaded areas represent a 95% credible interval for estimates of life-years gained for each strategy. All efficient (i.e., nondominated) strategies are displayed in the graph with a label, and those are the ones that save the most life-years while using fewer colonoscopies. Strategies that are either dominated or extended dominated are omitted from this figure, except for strategies 45–75,10 (United States Preventive Task Force [USPSTF] recommended, represented by a circle if efficient or as a cross otherwise) and strategy 50–70,10 (American College of Physicians [ACP] recommended, represented by a square). A darker, shaded region underneath the frontiers encompasses a 3-day vertical distance to the frontier, which Knudsen et al.[12] used to select “nearly efficient” strategies. Each panel presents 1 model specification.

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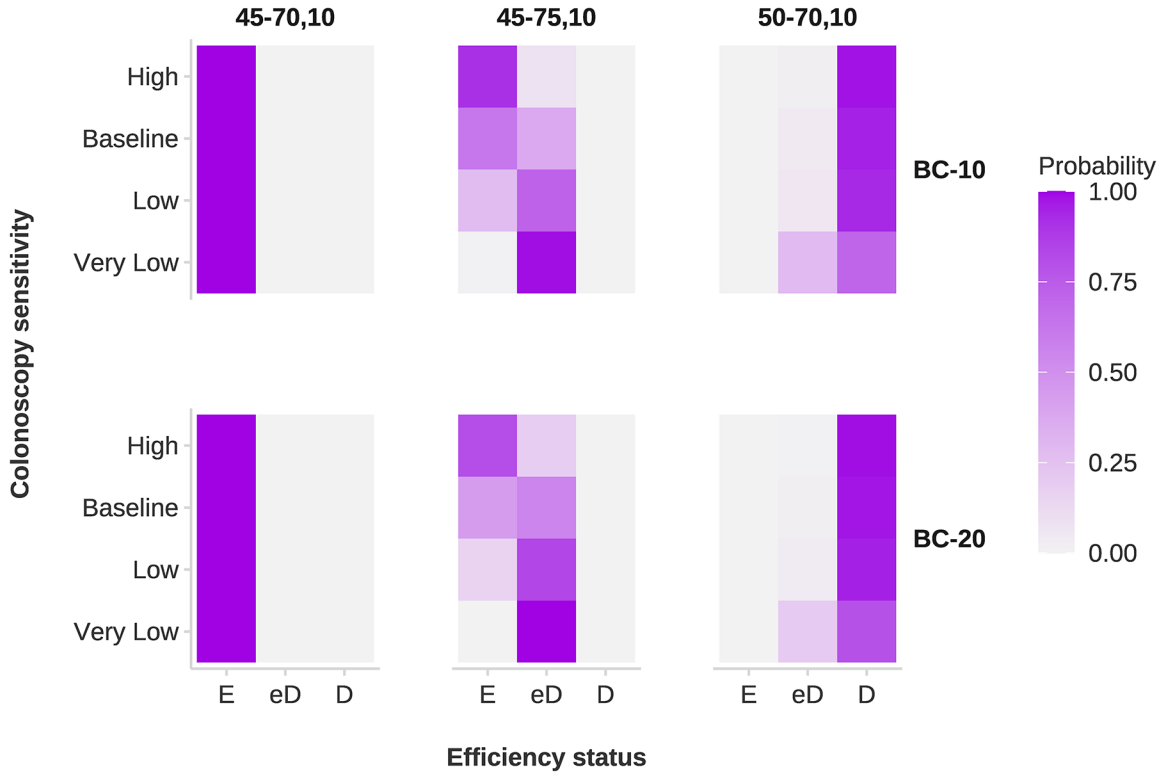


Figure 3.

Robustness of efficiency status by scenario for select strategies.

The horizontal axis represents the efficiency status of each strategy, coded as E (efficient), eD (extended dominated), and D (dominated). Colors represent the probability that each strategy will have each efficiency status. Each facet represents 1 colonoscopy screening strategy (columns) for each model specification (rows). The results illustrate that the United States Preventive Task Force (USPSTF)–recommended strategy is sometimes efficient but never dominated, whereas the American College of Physicians (ACP) strategy is almost always dominated and never efficient.

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Table 1

Outcomes under the Base-Case Scenario

Strategy	Efficiency Status	LYG/1,000	Colonoscopies/1,000	IBER
55–70,15	Efficient	274 (219, 353)	2,540 (2,430, 2,690)	—
50–70,10	Dominated	317 (254, 406)	3,410 (3,300, 3,550)	—
45–70,15	Efficient	324 (256, 419)	2,910 (2,750, 3,120)	7.53 (6.2, 9.4)
45–70,10	Efficient	343 (275, 440)	3,640 (3,510, 3,800)	38.5 (32.4, 46.1)
45–75,10	Extended dominated	347 (279, 444)	4,200 (4,110, 4,320)	—
45–70,5	Efficient	358 (290, 457)	5,520 (5,460, 5,580)	125 (98.6, 155)
45–75,5	Efficient	360 (291, 458)	6,010 (5,930, 6,100)	331 (245, 678)
45–80,5	Efficient	360 (292, 459)	6,410 (6,300, 6,520)	907 (420, 2,540)
45–85,5	Efficient	360 (292, 459)	6,700 (6,560, 6,820)	869,000 (−13,600, 21,300)

This table shows outcomes for the strategies found as efficient in the base-case scenario (model BC-20 and baseline colonoscopy sensitivity). In addition, strategies 45–75,10 (United States Preventive Task Force guideline-recommended strategy) and 50–70,10 (strategy representing American College of Physicians' guidelines) are included. Life-years gained (LYG) and colonoscopies are presented by 1,000 persons. The incremental burden-effectiveness ratio (IBER = incremental costs/incremental LYG) is high for the most intensive strategies since they only offer a small benefit and require more colonoscopies. All results are presented with 3 significant digits.

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